FAMILIAR WITH DIGITAL ELECTRONICS COMPONENT FOR DIGITAL LOGIC LAB

1st components

1) Register
2) Transistor
3) Bread Board
4) Switch
5) LED

Observations :

6) Digital IC (Gates)
7) Decorder
8) Power Supply
9) Wire
10) Digital clock

Register

→ A resistor is used to control the flow of electric current. In an LED circuit, the resistor is very important because it limits the current and prevents the LED from burning out. A resistor is a passive component that implements electrical resistance in circuits.

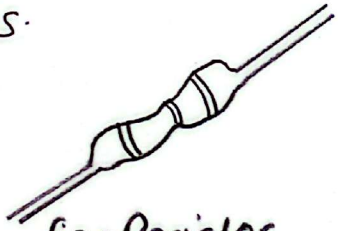


fig : Resistor

Transistor

→ A transistor act like an electronic switch or amplifies. In the circuit transistor helps to control the LED using a simple small current.

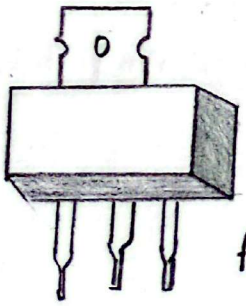


fig: Transistor

Bread Board

→ Bread board is used for building temporary circuits. It is useful for designers because it allows components to be removed & replaced. It is a temporary prototype.

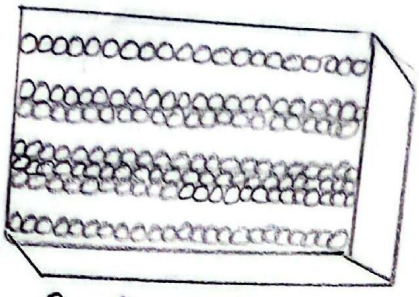


fig: Bread board

Switch

→ A switch is any device that is used to interrup the flow of electron in circuit and also connect other devices. multiple data cables and plugged into a switch to enable communication betn networked services.

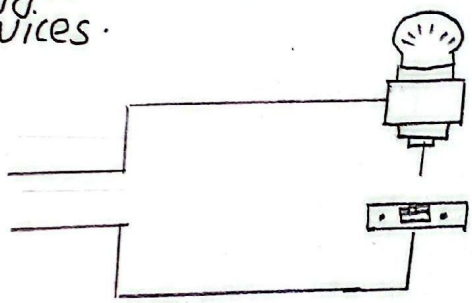


fig: switch

LED

→ LED light emmitting diods is a semi-conductor device which can emit light when a electronic current passes through it. To do this, holes from p-type semiconductor recombines with electron from n-type semi conductor.

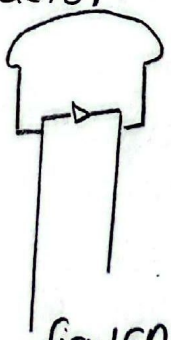


fig: LED

Digital IC's (gates)

→ It is a semiconductor based chip comprising on integrated set of digital circuit. It has functionality of logic AND or amplyfying of a signal.

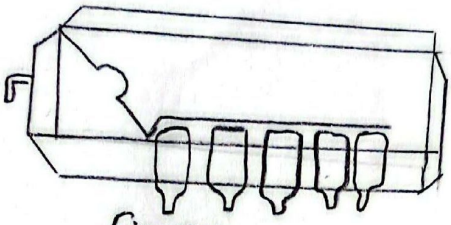
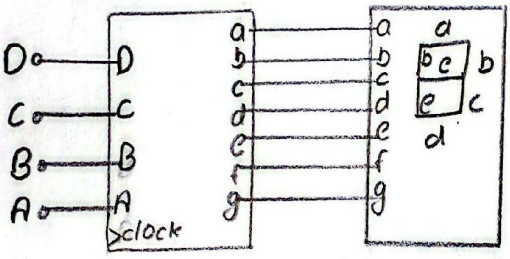


fig: IC's

Seven Segment decoder

→ A seven segment decoder is a combinational digital logic circuit that converts a coded decimal or BCD into a specific 7-bit output pattern



BCD-7segment decoder 7-segment display

Power Supply

→ A power Supply is an electrical device that supplies electric power to an electrical load. The main purpose of a power supply is to convert electric current from a source to the corect voltage.

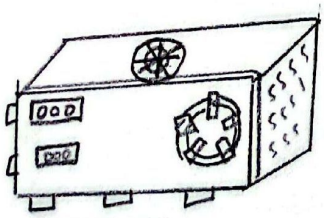


fig: power supply

Wire (Banana cable)

→ Wire banana cables, features banana plugs as connectors, and are mainly used for connecting devices.

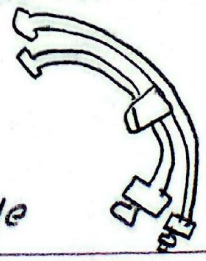


fig: Banana cable

Clock (digital)

→ A digital clock is a timekeeping device that represents time numerically using electronic digits rather than rotating hands on dials. Digital clock display time in discrete increments.

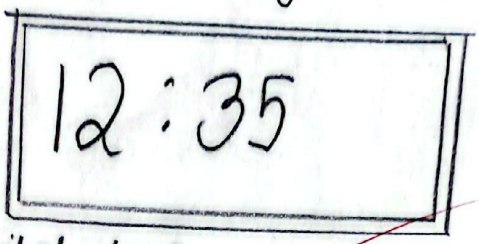


fig: digital clock

Digital trainer kit

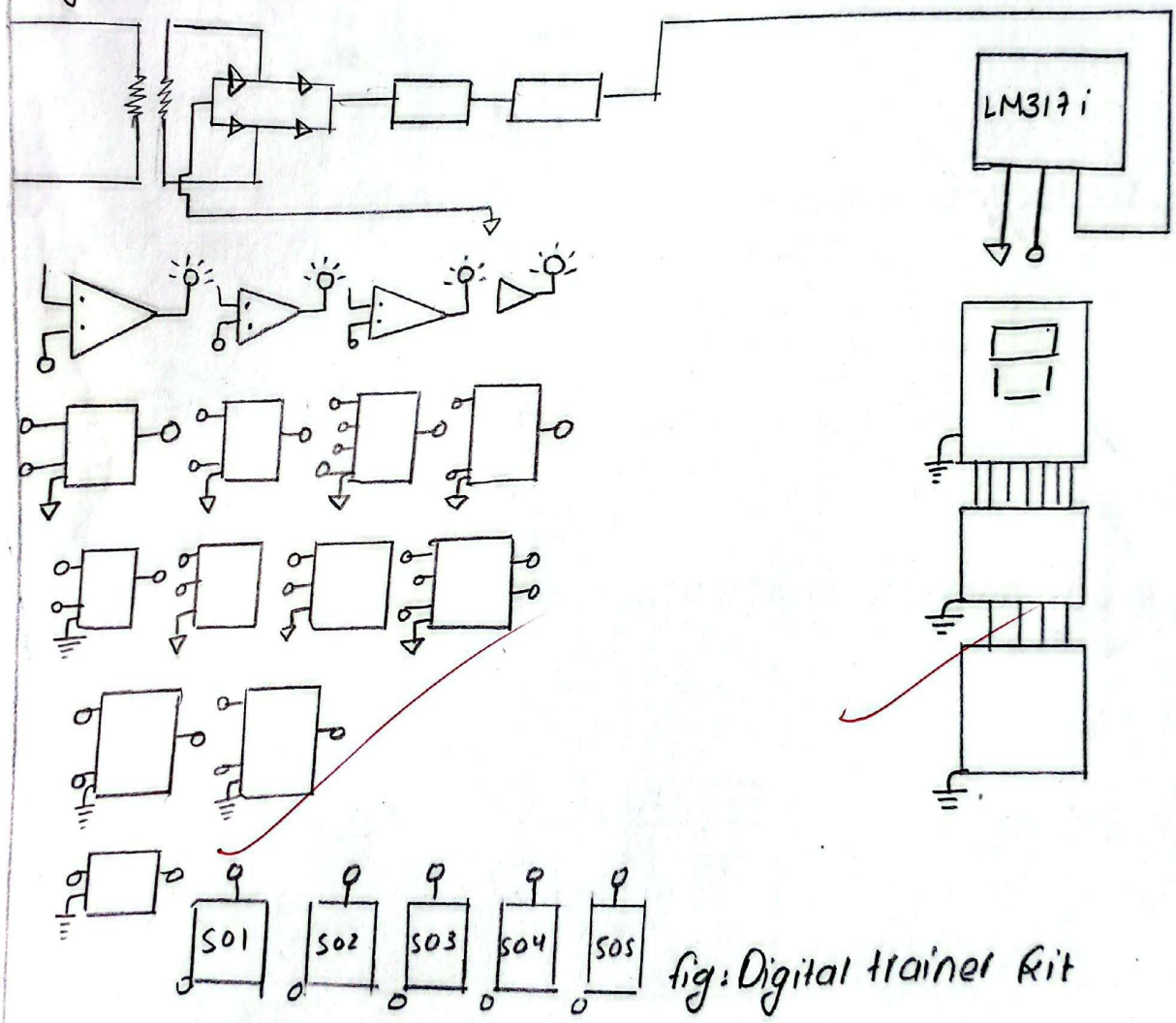
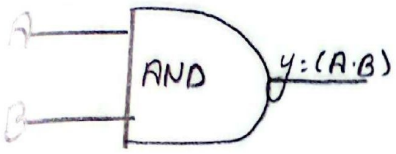


fig: Digital trainer kit

A digital trainer kit is an educational tool used to teach and expressions with the fundamental concepts of digital electronics. It helps to design, test and modify digital circuits.

Different Gates

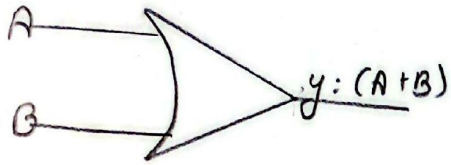
AND



A	B	y
1	1	1
1	0	0
0	1	0
0	0	0

→ AND gate is an electrical circuit that combines two signals so that the output is an if ~~et~~ both signals are present.

OR



A	B	y
1	1	1
1	0	1
0	1	1
0	0	0

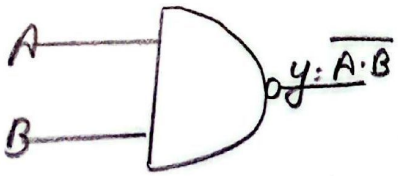
→ OR gate is a digital logic gate that gives an output of 1 when any of its inputs are 1 otherwis 0.

NOT

A	y
1	0
0	1

→ NOT gate is a digital logic gate which perform logic negation on its input.

NAND



A	B	A·B	$\overline{A\cdot B}$
0	1	1	0
0	0	0	1
1	1	0	1
1	0	0	1

→ A NAND gate is a digital combination logic unit/ circuit that performs logical inverse of AND gate.

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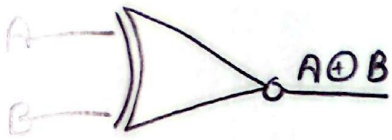
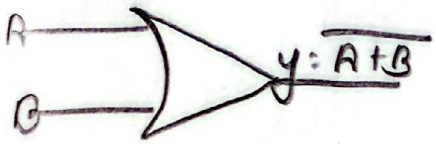
XOR



A	B	A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	1

A XOR is a type of logical Gates when has operants that result in a value of true if exactly, one has the value of true.

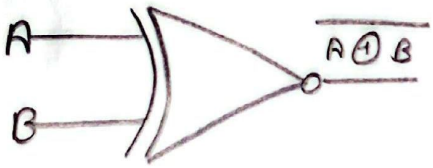
NOR



A	B	A+B	$\overline{A+B}$
1	1	1	0
1	0	1	0
0	1	1	0
0	0	0	1

NOR gate is a digital logic gate given that output of 0 when any of its input are 1.

XNOR



A	B	$A \overline{\oplus} B$
0	0	1
0	1	0
1	0	0
1	1	1

XNOR is a binary operator that operates in two logical inputs and produces a single logical output. Its IC no is 74266.

Conclusion

Thus, we successfully studied and became familiar with various digital electronic components. We also learned the practical working of logic gates by connecting there on a digital trainer and observing the output through a bulb (LED)

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